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FOURTH REVIEW UNDER THE EXTENDED CREDIT FACILITY—JOINT BANK-FUND STAFF DEBT SUSTAINABILITY ANALYSIS

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This low-income debt sustainability analysis (LIC DSA) updates the joint IMF/IDA DSA from February 14, 2013, reflecting the most recent macroeconomic developments. This updated DSA indicates that Burundi continues to be assessed at a high risk of debt distress. Both baseline public and external DSA suggest Burundi's public sector debt is sustainable given the current size and evolution of the debt stock. Compared to the 2012 DSA assessment, overall public debt sustainability improves modestly. Debt sustainability remains highly sensitive to shocks, due mainly to the narrow export base. The public DSA suggests that Burundi's overall public sector debt sustainability indicators are projected to improve in the medium and long run. However, the large downside risks and the vulnerability of the indicators to shocks point to the need for prudent fiscal and debt policies, and for structural reforms to promote private sector-led growth and exports diversification.[1]

[1] The DSA has been produced jointly by Bank and Fund staffs. The fiscal year for Burundi is January to December.

BACKGROUND

1. **The Debt Sustainability Analysis update indicates that Burundi continues to have a high risk of debt distress**. Nonetheless, the update of the debt sustainability analysis (DSA) using the uniform 5 percent discount rate in line with the recent Board decision shows some improvements.[1] While Burundi's assessment of high risk of debt distress remains, the PV of external debt-to-GDP and debt-to-government revenue ratios have improved and are now below their assessment thresholds (Figures 1 and 2 and Tables 1–4). The DSA update suggests that Burundi has limited borrowing space, underscoring that loans should continue to be highly concessional given its narrow export base.[2]

2. **Burundi is a weak policy performer for the purpose of determining the debt burden thresholds under the Debt Sustainability Framework (DSF).** Burundi's rating on the World Bank's Country Policy and Institutional Assessment (CPIA) has improved slightly in recent years. However, the performance is still low, and the average for the last three years—3.14 on a scale of 1 to 6—puts Burundi in the group of weak policy performer.[3]

3. **At end-2012, Burundi's public and publicly guaranteed external debt stood at US $487 million or 22.3 percent of GDP.** Burundi's stock of external debt has declined significantly since 2009 as a result of the debt relief under the enhanced Highly Indebted Poor Countries (HIPC) Initiative and the Multilateral Debt Relief Initiative (MDRI). About 90 percent of Burundi's outstanding nominal external PPG debt is owed to multilateral creditors, with bilateral creditors accounting for the remainder.

[1] On October 11, 2013, the Executive Boards of the IMF and the World Bank approved the reform of the discount rates presented in SM/12/271, resulting in a unique, unified 5 percent rate to be applied both to DSAs and to the calculation of the grant element (Board Decision N. 15462-(13–97)).

[2] Coffee and tea account for about 80 percent of exports.

[3] A score below 3.25 corresponds to a poor policy performance, according to the LIC Debt Sustainability Framework (DSF).

Text Table 1. Burundi: Stock External Debt, end-2012
(Millions of US dollars)

	Nominal	Percent of Total	Percent of GDP
Total Debt	487	100	22.3
Multilateral	440	90.4	20.1
Bilateral	46	9.6	2.1
Paris club	0	0.0	0.0
Non-Paris club	46	9.6	2.1
Commercial	0	0.0	0.0

Sources: Burundi authorities; and Bank-Fund staff estimates.

UNDERLYING ASSUMPTIONS

4. **While growth estimates remained at the levels projected in the 2012 DSA, other macroeconomic developments in 2012–13 underperformed the previous estimates and projections, mainly due to external shocks.** Trade deficit exceeded initial projections by about 3 percentage points of GDP in 2012 and 5 percentage points in 2013, as the terms of trade deteriorated by about 38 percent cumulatively in 2012–13. Coffee, which accounts for about two-thirds of exports, experienced a decline in international prices by about 30 percent during 2012–13. The import coverage of gross international reserves remained broadly unchanged from previous projections. Inflation is slightly higher than projections for 2013 mostly owing to the domestic supply shocks. The overall budget deficit widened by 1.3 and 0.3 percentage points of GDP compared to the 2012 DSA mainly owing to lower grants and weaker revenue performance.

5. **The macroeconomic outlook has been revised accordingly.** The average medium-term (2013–18) GDP growth is revised slightly downward compared to the 2012 DSA, although the long-term growth is broadly kept unchanged as the growth outlook remains unaltered. Due to this revision, *i.e.* lower nominal GDP base in the medium term, the size of the economy is also projected to be smaller in the long run, implying a lower level of sustainable debt. As the decline in coffee prices is projected to continue in 2014, medium-term exports growth was also revised downward.

Prices are expected to bottom out towards the middle of the decade; thus, long-run growth is kept broadly unchanged compared to the 2012 DSA.[4] Combined with oil price projections, which are higher, in the near term, than those in the 2012 DSA, the trade deficit is projected to be more pronounced in the medium term. Central government revenue projections have also been adjusted downward reflecting the permanent losses induced by the 2013 new income law and negative impact of the various shocks to economic activities. Financial assistance from donors, all types considered, is assumed to decline from about 20 percent of GDP in 2012 to 13 percent in the long run.[5]

6. Risks to the macroeconomic outlook stem mostly from the fragile social and security situation and the external environment. The protracted Euro Area debt crisis and the decelerating economic growth in emerging markets are likely to engender negative spillovers through the trade and investment channels. Uncertainty in donor support also poses risks. Despite the projected easing of oil and food international prices, uncertainty remains. Moreover, the recent influx of refugees continues to weigh heavily on the social and economic situation. Other socio-political developments are highly unpredictable in the run-up to the 2015 elections.

EXTERNAL DSA

7. **Under the baseline scenario, one indicator breaches the policy threshold during the medium term**. The PV of debt-to-exports ratio, although gradually declining, is projected to stay above the 100 percent policy threshold until around 2020. The debt service-to-exports ratio slightly and temporarily breaches the threshold. These projected developments are mostly due to Burundi's narrow export base and the relatively limited export potential at this time. In contrast, the PV of debt-to-GDP ratio, the PV of debt-to-revenues ratio, and the debt service-to-revenue ratio are expected to remain well below the indicative policy dependent thresholds throughout the projection period. Moreover, those indicators are somewhat stabilizing in the medium term and show a declining trend in the long run, indicating an improvement of the debt sustainability profile in the long run (Text Table 2, Figure 1 and Table 1). This stems from the intention of the authorities to

[4] In the medium and long terms, coffee prices are assumed to increase by 5 percent per year (which corresponds broadly to the average of the last two decades). Also, coffee production is assumed to expand by 5 percent per year, reflecting the expected outcome of the on-going reforms in the sector.

[5] Financial assistance from donors includes budget support, project loans and grants, humanitarian assistance, technical assistance, and financing related to elections and regional conflicts.

pursue sound macroeconomic and prudent debt policies. The combination of such policies is expected to alleviate debt burden indicators. The reduction of debt burden is a key pillar of the program currently implemented by the government and supported under the IMF's Extended Credit Facility (ECF).

Text Table 2. Burundi: Summary of Baseline External Debt Sustainability Indicators
(percent)

	Indicative Thresholds	2013	2023	2033
PV of debt to GDP	30	13.4	5.8	2.7
PV of debt to exports	100	248.5	78.4	32.0
PV of debt to revenue	200	101.9	44.3	20.9
Debt service to exports	15	8.3	5.1	2.2
Debt service to revenues	25	3.4	2.9	1.4

Sources: Burundi authorities; and Bank-Fund staff estimates.

8. **Alternative scenarios and stress tests highlight the high vulnerability of the debt sustainability profile to adverse shocks.** Under a scenario of combined adverse shocks on GDP growth, exports, and FDI flow, the debt indicators worsen significantly compared to the baseline scenario; four of the debt indicators breach the threshold in the medium term and return broadly close to the baseline in the long run.[6] However, under a scenario that assumes continuation of policies during the last ten years, two indicators breach the threshold; most indicators would double compared to that under the baseline scenario and would not improve even in the long run.[7] These results underscore the need to foster a sound macroeconomic environment that would promote

[6] The combination of shocks assumes that, during 2014–15, GDP growth, export growth, USD GDP deflator and non-debt creating flows will be at their historical averages minus one-half standard deviation.

[7] The historical scenario assumes that, throughout the projection period, key macroeconomic variables will stay at their respective average during the last ten years. The some economic variables in 2009 were adjusted as Burundi benefited from the Heavily Indebted Poor Countries Initiative and the Multilateral Debt Relief Initiative during that year.

growth, export diversification, and inflow of foreign direct investment, and to continue the reform measures to avoid returning to policies in the past.[8]

9. **All scenarios suggest that Burundi's narrow export base is the most significant factor that contributes to the vulnerability of Burundi's debt sustainability.** The PV of debt-to-exports ratio remains above the policy threshold of 100 percent in the baseline, historical, and stress tests scenarios. Although, the trend declines in the long term, the ratio stays stubbornly high, particularly under the historical scenario.

PUBLIC DSA

10. **Public debt indicators, including external and domestic, are expected to gradually improve under the baseline scenario**. The improvement is due primarily to a decline in the public sector borrowing requirement, reflecting the widening of the revenue base and the gradual decline in government spending in the post reconstruction period. The ratios of the PV of public debt to GDP and public debt to revenues remain low, reflecting Burundi's reliance on grants and highly concessional loans to finance reconstruction and poverty reduction.

11. **However, public debt indicators are highly vulnerable to shocks.** Under a shock scenario that combines a lower GDP growth and a larger primary deficit, the PV of debt-to-GDP ratio is projected to enlarge by 6 percentage points (above the baseline scenario) throughout the projection period, and the PV of debt-to-revenue ratio by about 20 percentage points.[9] These results underscore the need for prudent fiscal policy and avoidance of past unsustainable borrowing policies. A swift adoption and implementation of a strategy based on the recent Debt Management Performance Assessment (DeMPA) would be crucial. The debt service to revenue ratio is not significantly affected by alternative scenarios and shocks because additional borrowing is expected to be on highly concessional terms.

[8] In the event the assumption on coffee production does not materialize and the country falls back into a fragility trap, the debt indicators would significantly worsen.

[9] The scenario assumes that, in 2013-14, GDP growth and primary balance will be at their historical average minus one-half standard deviation.

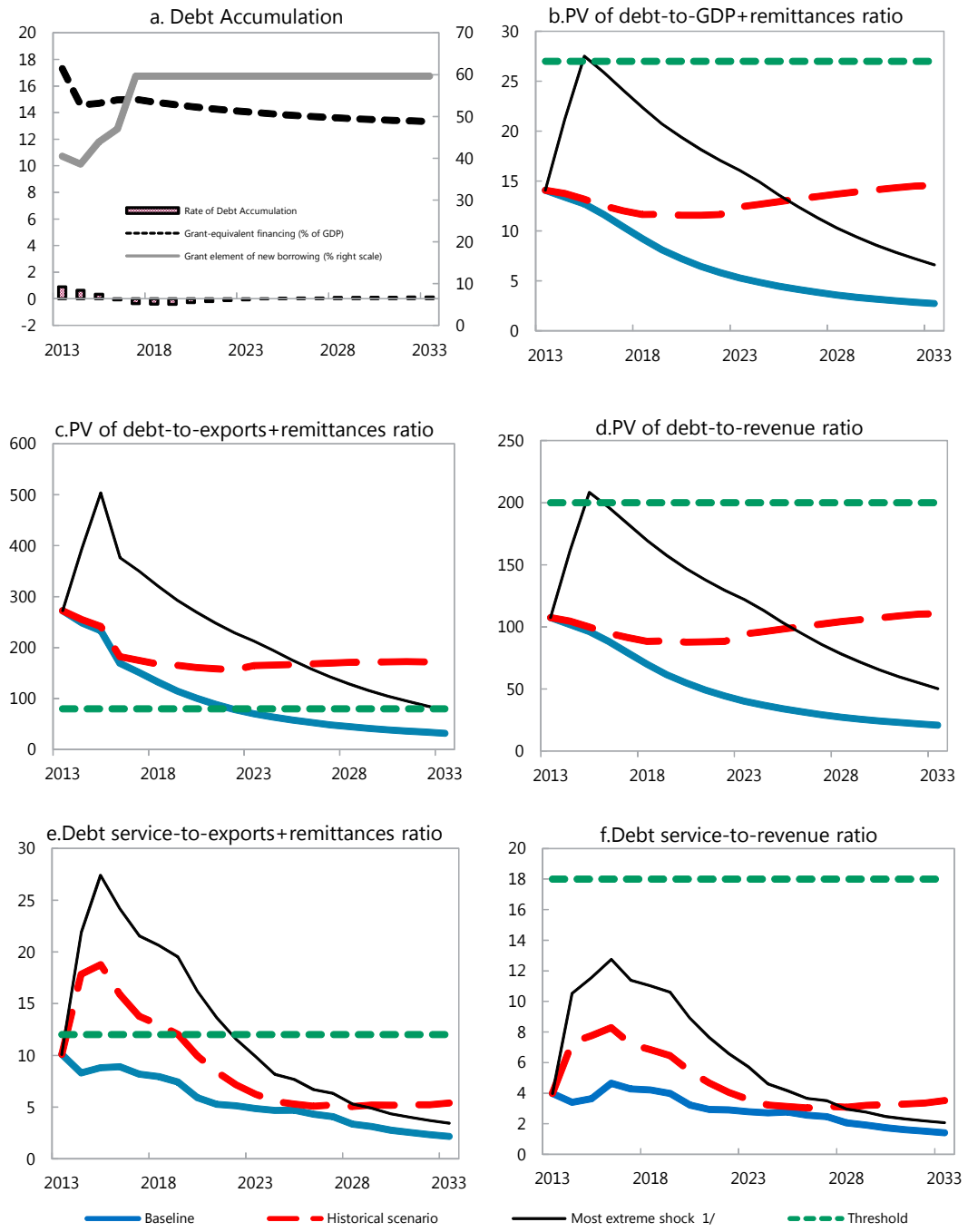
CONCLUSION

12. **Based on this LIC-DSA, staffs are of the view that Burundi continues to face a high risk of debt distress**. The debt sustainability indicators improved slightly compared to the 2012 DSA. However, the classification remains unchanged, considering that, as in the 2012 DSA, the PV of debt-to-export ratio remains above the policy threshold under the baseline.

13. **Based on this high risk classification and on the vulnerabilities shown through the alternative and stress tests scenarios, Burundi should pursue sound macroeconomic and prudent debt policies.** In particular, the analysis points to the importance of enlarging export base and diversifying export markets. This would include swift implementation of reforms in the coffee sector and unlocking export potential in other sectors (mining, tea, horticulture, and tourism). It would also be essential to continue sound policies as policy reversals are shown in the analysis as having serious hindering effect on debt sustainability. Finally, given the high risk of debt distress and the vulnerabilities, staffs encourage the authorities to continue to seek maximum concessionality in their external financing, with all nonconcessional borrowing regularly reviewed, monitored, and reported to ensure full transparency and sound governance. Staffs encourage the authorities to finalize the new law on public debt, which would provide an overreaching debt legal framework and help determine the objective, the strategy, the signing authority, and other aspects of debt management. The strengthening of debt management practices now underway is a good step towards reinforcing debt sustainability. Staffs encourage the authorities to expedite the implementation of the World Bank DeMPA mission assessment to facilitate putting in place a comprehensive medium-term debt strategy.

The authorities broadly share staffs' assessment

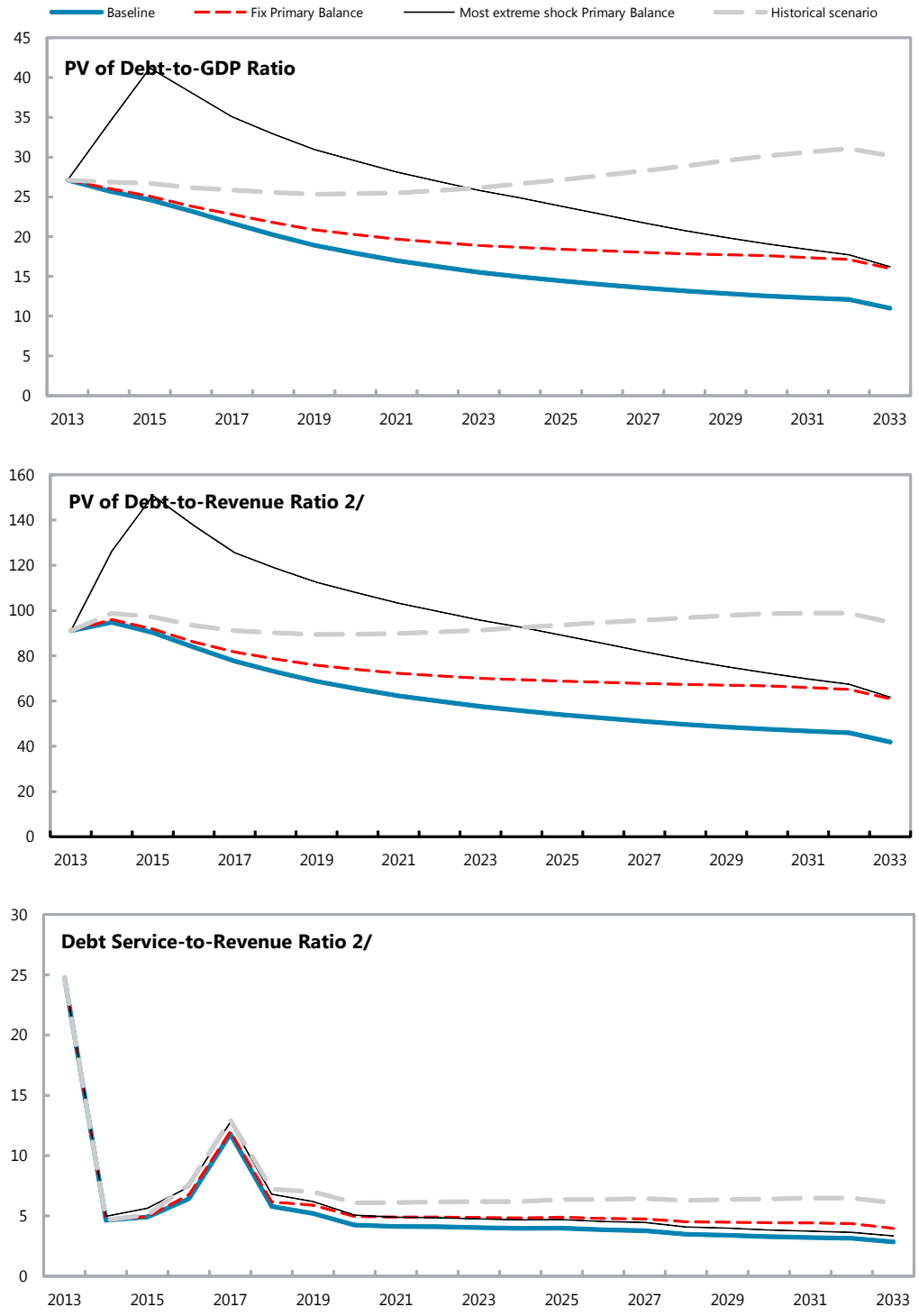
Figure 1. Burundi: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2013-2033 1/



Sources: Burundi authorities; and IMF staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2023. In figure b. it corresponds to a Non-debt flows shock; in c. to a Non-debt flows shock; in d. to a Non-debt flows shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock.

Figure 2.Burundi: Indicators of Public Debt Under Alternative Scenarios, 2013–2033 1/



Sources: Burundi authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2023.
2/ Revenues are defined inclusive of grants.

Table 1.: External Debt Sustainability Framework, Baseline Scenario, 2010-2033[1/]

(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average [6/]	Standard Deviation [6/]	Projections									
	2010	2011	2012			2013	2014	2015	2016	2017	2018	2013-2018 Average	2023	2033	2019-2033 Average
External debt (nominal) 1/	**22.5**	**23.4**	**22.3**			**18.5**	**17.4**	**16.6**	**15.3**	**13.7**	**12.2**		**7.5**	**4.7**	
of which: public and publicly guaranteed (PPG)	22.5	23.4	22.3			18.5	17.4	16.6	15.3	13.7	12.2		7.5	4.7	
Change in external debt	1.3	0.9	-1.1			-3.8	-1.0	-0.8	-1.3	-1.6	-1.5		-0.6	-0.1	
Identified net debt-creating flows	-0.4	2.3	0.5			-0.1	-0.2	-0.4	-0.8	-1.4	-1.5		-2.8	-1.4	
Non-interest current account deficit	**12.2**	**14.6**	**18.4**	**7.9**	**7.8**	**22.9**	**21.2**	**21.0**	**19.9**	**17.9**	**17.2**		**15.1**	**16.9**	17.1
Deficit in balance of goods and services	34.6	34.0	37.2			35.5	32.2	31.2	29.6	27.5	26.4		22.9	23.3	
Exports	8.9	10.3	9.3			5.2	5.4	5.5	6.9	6.9	7.0		7.5	8.6	
Imports	43.5	44.3	46.5			40.7	37.6	36.6	36.4	34.4	33.4		30.5	31.9	
Net current transfers (negative = inflow)	-23.0	-20.2	-19.0	-20.3	4.2	-12.8	-11.2	-10.3	-9.7	-9.5	-9.2		-7.7	-6.4	-7.3
of which: official	-17.3	-12.8	-13.8			-7.9	-6.6	-5.9	-5.4	-5.4	-5.2		-4.4	-3.6	
Other current account flows (negative = net inflow)	0.5	0.7	0.3			0.3	0.2	0.1	0.1	0.0	-0.1		-0.2	0.0	
Net FDI (negative = inflow)	**-10.0**	**-10.6**	**-16.6**	**-6.1**	**7.3**	**-22.3**	**-20.8**	**-20.8**	**-20.1**	**-18.8**	**-18.1**		**-17.5**	**-18.1**	-18.4
Endogenous debt dynamics 2/	**-2.6**	**-1.6**	**-1.3**			**-0.7**	**-0.7**	**-0.6**	**-0.6**	**-0.6**	**-0.5**		**-0.4**	**-0.3**	
Denominator: 1+g+r+gr	1.1	1.1	1.1			1.2	1.1	1.1	1.1	1.1	1.1		1.1	1.1	
Contribution from nominal interest rate	0.1	0.1	0.1			0.1	0.1	0.1	0.1	0.1	0.1		0.1	0.0	
Contribution from real GDP growth	-0.9	-0.9	-0.9			-0.9	-0.8	-0.8	-0.8	-0.7	-0.7		-0.4	-0.3	
Contribution from price and exchange rate changes	-1.7	-0.8	-0.6			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**1.7**	**-1.4**	**-1.6**			**-3.7**	**-0.8**	**-0.5**	**-0.6**	**-0.2**	**0.0**		**2.1**	**1.3**	
of which: exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	16.9			14.1	13.4	12.7	11.6	10.4	9.2		5.3	2.7	
In percent of exports	...	...	180.7			272.6	248.5	233.0	169.2	151.2	132.3		70.1	32.0	
PV of PPG external debt	**...**	**...**	**16.9**			**14.1**	**13.4**	**12.7**	**11.6**	**10.4**	**9.2**		**5.3**	**2.7**	
In percent of exports	**...**	**...**	**180.7**			**272.6**	**248.5**	**233.0**	**169.2**	**151.2**	**132.3**		**70.1**	**32.0**	
In percent of government revenues	**...**	**...**	**108.1**			**107.5**	**101.9**	**96.4**	**88.3**	**78.9**	**69.9**		**40.2**	**20.9**	
Debt service-to-exports ratio (in percent)	**1.3**	**2.2**	**5.3**			**10.1**	**8.3**	**8.8**	**8.9**	**8.2**	**8.0**		**4.9**	**2.2**	
PPG debt service-to-exports ratio (in percent)	**1.3**	**2.2**	**5.3**			**10.1**	**8.3**	**8.8**	**8.9**	**8.2**	**8.0**		**4.9**	**2.2**	
PPG debt service-to-revenue ratio (in percent)	**0.8**	**1.4**	**3.2**			**4.0**	**3.4**	**3.6**	**4.6**	**4.3**	**4.2**		**2.8**	**1.4**	
Total gross financing need (Millions of U.S. dollars)	46.8	91.2	55.0			30.2	27.3	24.7	15.9	-11.2	-17.1		-125.5	-137.2	
Non-interest current account deficit that stabilizes debt ratio	10.9	13.6	19.5			26.7	22.3	21.8	21.3	19.5	18.7		15.7	17.1	
Key macroeconomic assumptions															
Real GDP growth (in percent)	5.1	4.2	4.0	4.1	0.9	4.5	4.7	4.8	5.0	5.2	5.4	4.9	5.9	7.0	6.4
GDP deflator in US dollar terms (change in percent)	8.8	3.9	2.5	8.4	8.5	12.1	4.9	3.1	3.0	2.9	3.0	4.8	2.5	2.3	2.4
Effective interest rate (percent) 5/	0.3	0.6	0.5	0.7	0.2	0.7	0.8	0.9	1.0	1.0	1.1	0.9	1.1	0.8	1.0
Growth of exports of G&S (US dollar terms, in percent)	52.3	25.1	-3.1	21.5	25.6	-35.1	14.6	9.5	36.2	8.3	9.9	7.2	10.0	11.4	10.4
Growth of imports of G&S (US dollar terms, in percent)	76.0	10.2	12.0	26.6	32.2	2.4	1.7	5.2	7.5	2.1	5.4	4.1	6.0	8.4	8.6
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	40.5	38.6	43.9	47.0	59.6	59.6	48.2	59.6	59.6	59.6
Government revenues (excluding grants, in percent of GDP)	14.5	16.5	15.6			13.1	13.1	13.2	13.2	13.2	13.2		13.2	13.2	13.2
Aid flows (in Millions of US dollars) 7/	490.9	513.5	467.9			474.0	437.0	481.4	530.4	578.5	619.3		878.1	1995.6	
of which: Grants	460.9	487.7	423.0			456.3	422.1	459.3	510.4	561.7	601.0		850.9	1930.6	
of which: Concessional loans	30.0	25.8	44.9			17.7	14.9	22.1	20.1	16.8	18.2		27.1	65.0	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			17.3	14.6	14.7	15.0	15.0	14.8		14.1	13.3	13.8
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			94.7	94.1	95.0	96.6	98.8	98.8		98.8	98.7	98.7
Memorandum items:															
Nominal GDP (Millions of US dollars)	2029.2	2196.1	2341.8			2743.3	3015.2	3256.5	3521.2	3810.1	4134.9		6166.2	14773.0	
Nominal dollar GDP growth	14.3	8.2	6.6			17.1	9.9	8.0	8.1	8.2	8.5	10.0	8.6	9.5	8.9
PV of PPG external debt (in Millions of US dollars)			368.3			389.3	406.7	416.7	413.8	400.5	384.9		329.8	410.2	
(PVt-PVt-1)/GDPt-1 (in percent)						0.9	0.6	0.3	-0.1	-0.4	-0.4	0.2	-0.1	0.1	0.0
Gross workers' remittances (Millions of US dollars)	...	...	...			...	...	...	...	...	...		...	...	
PV of PPG external debt (in percent of GDP + remittances)	...	...	16.9			14.1	13.4	12.7	11.6	10.4	9.2		5.3	2.7	
PV of PPG external debt (in percent of exports + remittances)	...	...	180.7			272.6	248.5	233.0	169.2	151.2	132.3		70.1	32.0	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	5.3			10.1	8.3	8.8	8.9	8.2	8.0		4.9	2.2	

Sources: Burundi authorities; and IMF staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2. Burundi: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013–2033

(In percent)

	Projections							
	2013	2014	2015	2016	2017	2018	**2023**	2033
PV of debt-to GDP ratio								
Baseline	15	14	13	12	11	9	**5**	3
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	15	14	13	13	12	12	**12**	15
A2. New public sector loans on less favorable terms in 2013-2033 2	15	14	14	13	12	11	**7**	5
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	15	14	14	12	11	10	**6**	3
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	15	14	14	13	12	10	**6**	3
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	15	15	14	13	12	10	**6**	3
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	15	22	28	26	24	23	**16**	7
B5. Combination of B1-B4 using one-half standard deviation shocks	15	20	24	22	21	19	**13**	6
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	15	19	18	17	15	13	**8**	4
PV of debt-to-exports ratio								
Baseline	281	256	239	174	155	136	**72**	32
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	281	262	247	186	178	170	**165**	170
A2. New public sector loans on less favorable terms in 2013-2033 2	281	263	252	188	171	152	**93**	55
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	281	257	241	175	157	137	**72**	33
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	281	317	354	259	233	205	**113**	49
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	281	257	241	175	157	137	**72**	33
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	281	400	510	381	354	324	**214**	77
B5. Combination of B1-B4 using one-half standard deviation shocks	281	377	454	338	312	283	**182**	68
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	281	257	241	175	157	137	**72**	33
PV of debt-to-revenue ratio								
Baseline	111	105	99	90	81	72	**41**	21
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	111	107	102	97	93	90	**95**	111
A2. New public sector loans on less favorable terms in 2013-2033 2	111	108	104	98	89	81	**53**	36
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	111	107	102	94	84	74	**43**	22
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	111	109	107	99	89	79	**47**	23
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	111	111	108	99	89	78	**45**	23
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	111	164	211	199	185	171	**123**	50
B5. Combination of B1-B4 using one-half standard deviation shocks	111	149	180	169	156	143	**100**	42
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	111	148	139	128	114	101	**58**	30

Table 2. Burundi: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013-2033 (concluded)

(In percent)

Debt service-to-exports ratio								
Baseline	10	8	9	9	8	8	**5**	2
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	10	18	19	16	14	13	**6**	5
A2. New public sector loans on less favorable terms in 2013-2033 2	10	19	21	18	17	16	**7**	3
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	10	19	21	18	16	15	**7**	2
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	10	22	28	25	22	21	**10**	4
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	10	19	21	18	16	15	**7**	2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	10	19	23	22	19	18	**9**	6
B5. Combination of B1-B4 using one-half standard deviation shocks	10	20	24	22	19	18	**9**	5
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	10	19	21	18	16	15	**7**	2
Debt service-to-revenue ratio								
Baseline	4	3	4	5	4	4	**3**	1
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	4	7	8	8	7	7	**4**	4
A2. New public sector loans on less favorable terms in 2013-2033 2	4	8	9	10	9	9	**4**	2
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	4	8	9	10	9	8	**4**	2
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	4	8	9	9	8	8	**4**	2
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	4	8	9	10	9	9	**5**	2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	4	8	10	11	10	10	**5**	4
B5. Combination of B1-B4 using one-half standard deviation shocks	4	8	9	11	10	9	**5**	3
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	4	11	12	13	12	11	**6**	2
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	55	55	55	55	55	55	**55**	55

Sources: Burundi authorities; and IMF staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 3. Burundi: Public Sector Debt Sustainability Framework, Baseline Scenario, 2010-2033

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2010	2011	2012	Average 5/	Standard Deviation 5/	2013	2014	2015	2016	2017	2018	2013-18 Average	2023	2033	2019-33 Average
Public sector debt 1/	39.7	39.6	37.7			31.5	29.8	28.5	26.8	25.0	23.2		17.7	12.9	
of which: foreign-currency denominated	22.5	23.4	22.3			18.5	17.4	16.6	15.3	13.7	12.2		7.5	4.7	
Change in public sector debt	14.7	-0.1	-1.9			-6.2	-1.7	-1.2	-1.7	-1.9	-1.8		-0.8	-1.1	
Identified debt-creating flows	-0.2	1.6	-1.1			-5.8	-1.7	-0.6	-0.8	-1.1	-0.9		-0.6	-0.1	
Primary deficit	2.3	2.5	3.0	-3.8	14.6	1.3	0.7	1.1	0.9	0.3	0.3	0.8	0.3	0.7	0.4
Revenue and grants	37.3	38.7	33.7			29.7	27.1	27.3	27.7	27.9	27.7		27.0	26.2	
of which: grants	22.7	22.2	18.1			16.6	14.0	14.1	14.5	14.7	14.5		13.8	13.1	
Primary (noninterest) expenditure	39.6	41.2	36.7			31.1	27.8	28.4	28.6	28.2	28.1		27.3	27.0	
Automatic debt dynamics	-2.5	-0.8	-3.7			-7.0	-2.3	-1.7	-1.7	-1.4	-1.3		-0.9	-0.8	
Contribution from interest rate/growth differential	-2.6	-4.5	-6.2			-5.6	-2.8	-2.2	-2.0	-1.7	-1.6		-1.0	-0.9	
of which: contribution from average real interest rate	-1.4	-2.9	-4.7			-3.9	-1.4	-0.9	-0.6	-0.4	-0.3		0.0	0.0	
of which: contribution from real GDP growth	-1.2	-1.6	-1.5			-1.6	-1.4	-1.4	-1.4	-1.3	-1.3		-1.0	-0.9	
Contribution from real exchange rate depreciation	0.1	3.6	2.5			-1.5	0.5	0.6	0.3	0.3	0.3		...	...	
Other identified debt-creating flows	0.0	0.0	-0.4			-0.1	-0.1	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	-0.4			-0.1	-0.1	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	14.9	-1.7	-0.9			-0.5	0.0	-0.6	-0.9	-0.8	-0.8		-0.2	-1.0	
Other Sustainability Indicators															
PV of public sector debt	...	...	32.3			27.1	25.7	24.6	23.2	21.7	20.2		15.5	11.0	
of which: foreign-currency denominated	...	...	16.9			14.1	13.4	12.7	11.6	10.4	9.2		5.3	2.7	
of which: external	...	...	16.9			14.1	13.4	12.7	11.6	10.4	9.2		5.3	2.7	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	5.7	11.3	18.2			10.9	6.4	7.2	7.9	9.5	9.8		9.9	9.5	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	96.0			91.1	94.8	90.2	83.8	77.6	72.9		57.5	41.9	
PV of public sector debt-to-revenue ratio (in percent)	...	...	207.2			206.8	195.8	186.6	175.9	164.4	153.3		117.8	83.4	
of which: external 3/	...	...	108.1			107.5	101.9	96.4	88.3	78.9	69.9		40.2	20.9	
Debt service-to-revenue and grants ratio (in percent) 4/	1.8	12.5	36.1			24.8	4.6	4.9	6.4	11.8	5.8		4.0	2.8	
Debt service-to-revenue ratio (in percent) 4/	4.5	29.4	77.9			56.2	9.6	10.1	13.5	25.0	12.2		8.3	5.7	
Primary deficit that stabilizes the debt-to-GDP ratio	-12.4	2.6	5.0			7.6	2.4	2.3	2.6	2.2	2.1		1.1	1.8	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	5.1	4.2	4.0	4.1	0.9	4.5	4.7	4.8	5.0	5.2	5.4	4.9	5.9	7.0	6.4
Average nominal interest rate on forex debt (in percent)	0.3	0.6	0.5	0.7	0.2	0.7	0.8	0.9	1.0	1.0	1.1	0.9	1.1	0.8	1.0
Average real interest rate on domestic debt (in percent)	6.4	-1.2	-10.3	-3.3	8.5	-12.4	-1.7	0.1	0.6	1.8	2.4	-1.5	2.7	2.3	2.8
Real exchange rate depreciation (in percent, + indicates depreciation)	0.3	19.1	12.7	3.7	12.2	-7.7	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	8.9	6.4	17.3	13.3	9.5	20.7	8.1	6.4	6.0	5.4	5.5	8.7	4.8	4.4	4.6
Growth of real primary spending (deflated by GDP deflator, in percent)	0.1	0.1	-0.1	0.1	0.2	-0.1	-0.1	0.1	0.1	0.0	0.0	0.0	0.1	0.1	0.1
Grant element of new external borrowing (in percent)	...	...	...	...	...	40.5	38.6	43.9	47.0	59.6	59.6	48.2	59.6	59.6	...

Sources: Burundi authorities; and staff estimates and projections.

1/ Central government gross debt is used.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 4. Burundi: Sensitivity Analysis for Key Indicators of Public Debt 2013-2033

	Projections							
	2013	2014	2015	2016	2017	2018	2023	2033
PV of Debt-to-GDP Ratio								
Baseline	27	26	25	23	22	20	16	11
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	27	27	27	26	26	26	26	30
A2. Primary balance is unchanged from 2013	27	26	25	24	23	22	19	16
A3. Permanently lower GDP growth 1/	27	26	25	23	22	21	16	13
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	27	26	26	24	23	22	17	13
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	27	34	41	38	35	33	26	16
B3. Combination of B1-B2 using one half standard deviation shocks	27	31	34	32	30	28	22	16
B4. One-time 30 percent real depreciation in 2014	27	31	30	28	26	24	18	12
B5. 10 percent of GDP increase in other debt-creating flows in 2014	27	31	30	27	26	24	19	13
PV of Debt-to-Revenue Ratio 2/								
Baseline	91	95	90	84	78	73	58	42
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	91	99	97	93	91	90	91	95
A2. Primary balance is unchanged from 2013	91	96	92	86	82	79	70	61
A3. Permanently lower GDP growth 1/	91	95	91	84	78	74	60	49
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	91	96	93	87	81	77	64	51
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	91	126	151	138	126	119	96	62
B3. Combination of B1-B2 using one half standard deviation shocks	91	113	124	115	106	100	82	59
B4. One-time 30 percent real depreciation in 2014	91	116	109	101	93	87	67	44
B5. 10 percent of GDP increase in other debt-creating flows in 2014	91	114	108	99	92	87	69	48
Debt Service-to-Revenue Ratio 2/								
Baseline	25	5	5	6	12	6	4	3
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	25	5	5	8	13	7	6	6
A2. Primary balance is unchanged from 2013	25	5	5	7	12	6	5	4
A3. Permanently lower GDP growth 1/	25	5	5	6	12	6	4	3
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	25	5	5	7	12	6	4	3
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	25	5	6	16	21	8	5	4
B3. Combination of B1-B2 using one half standard deviation shocks	25	5	6	12	17	7	5	4
B4. One-time 30 percent real depreciation in 2014	25	5	6	7	13	7	5	3
B5. 10 percent of GDP increase in other debt-creating flows in 2014	25	5	6	12	12	7	4	3

Sources: Burundi authorities; and staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.